

Problem Description:

You are given an array A of Q integers and Q queries. In each query, you are given an integer i ($1 \leq i \leq N$).

Your task is to find the minimum index greater than i ($1 \leq i \leq N$) such that:

1. Sum of digits of A_i is greater than the sum of digits of A_j
2. $A_i < A_j$

If there is no answer, then print -1.

Constraints

$$1 \leq N, Q \leq 10^5$$

$$1 \leq A_i \leq 10^9$$

$$1 \leq Q_i \leq N$$

Input format

- The first line contains two numbers N and Q .
- The next line contains N numbers.
- Next Q lines contain Q queries.

Output format

Print the answer as described in the problem

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5 5
62 70 28 62 92
1
5
3
4
2

EXPECTED OUTPUT

2 -1 4 -1 -1

Test Case 2

INPUT (STDIN)

6 6
12 71 21 16 91
11
15
31
14
21

EXPECTED OUTPUT

-1 -1 -1 -1 -1 -1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(arr[x]<arr[y])

Test Case 2

KEYWORD

if(arr2[x]>arr2[y])

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

10

Test Case 2

TOKEN COUNT

310

Test Case 3

NLOC

57